CHA-2.A.1 The difference quotients $\frac{f(a+h)-f(a)}{h}$ and $\frac{f(x)-f(x)}{x-a}$ express the average rate of change of a function over an interval.

CHA-2.B.1 The instantaneous rate of change of a function at x = a can be expressed by $\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$ or $\lim_{x\to a}\frac{f(x)-f(x)}{x-a}$ provided the limit exists. These are equivalent forms of the definition of the derivative and are denoted $f'(a)$.

2-1 Average and Instantaneous Rates of Change

Get Ready…

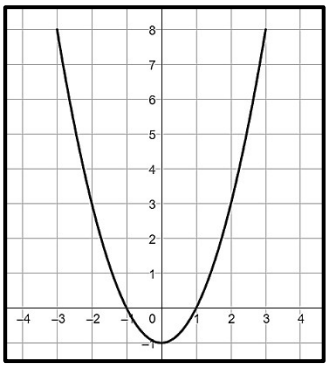
Let's revisit Average Rates of Change and Instantaneous Rates of Change (from sec 1-1).

Average Rate of Change (AROC)	The **average rate of change of a function** $f(x)$ over the interval $[a,b]$ is given by: $$AROC = \frac{f(b)-f(a)}{b-a}$$ The average rate of change tells us the slope of the line through the two points $(a, f(a))$ *and* $(b, f(b))$. This is also known as the slope of a **secant line**.

Find the average rate of change of the following functions over the interval $[-1,3]$.

A.) $f(x) = \sqrt{2x+3}$

B.)



We can calculate the AROC over any interval.

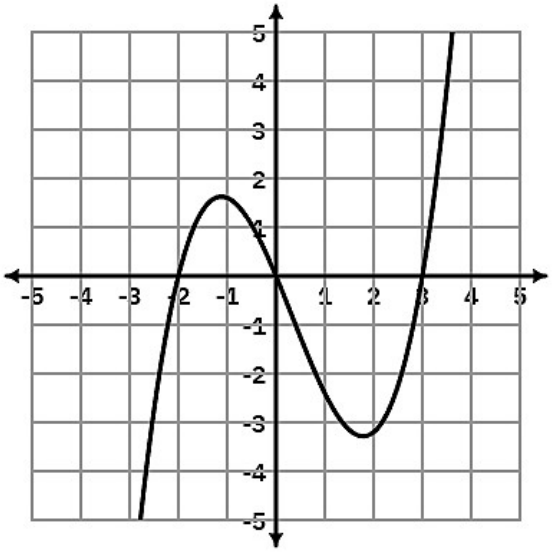
Example 1: Write an expression for the Average Rate of Change of $f(x)$ over the following intervals:

a.) $[a, x]$

b.) $[a, a+h]$

The **instantaneous rate of change of a function** at a point (a, b) is defined as the slope of the **tangent** line at the point (a, b).

Given the graph of $f(x)$ at the right, how can we find the instantaneous rate of change for $f(x)$ at $x = -2$?



Average Rate of Change (AROC)	The following quotients express the average rate of change of a function over an interval: $$\frac{f(x)-f(a)}{x-a} \quad \text{or} \quad \frac{f(a+h)-f(a)}{h}$$ This is also known as the slope of a **secant line**.

Instantaneous Rate of Change at a point (x = a)	The following limits express the **instantaneous rate of change** of a function $f(x)$ at $x = a$ $$\lim_{x\to a}\frac{f(x)-f(a)}{x-a} \quad \text{or} \quad \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$$ This is also known as the slope of the **tangent line at $x = a$**

Using the Limit Process to Compute Instantaneous Rate of Change

Example 2: Find the instantaneous rate of change of $f(x)$ at $x = 3$ given that $f(x) = 4x - 1$.

Example 3: Find the instantaneous rate of change of $g(x) = \sqrt{x}$ at $x = 4$.

Example 4: Find the equation of the tangent line to $g(x) = \sqrt{x}$ at $x = 4$.